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Study Case Data Analyst

Customer Churn Analytics & Retention Strategy

Presented by
Robert Gunanta Bukit

Business and Data Overview

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As a **digital financing** lending service provider, sustainable business growth depends on **user retention** and **the efficiency of acquisition channels**. As operational progress, the risk of customer churn either in form of **inactivity** or **no repeat borrowing** becomes a major challenges that has direct impact on the business.

01. Users
250 rows 5 columns

Column	Description
user_id	Customer identifier
signup_date	Registration date
acqstn_chnl	Acquisition channel
age	Customer age
city	Customer location

02. Activity
2,959 rows 4 columns

Column	Description
user_id	Customer identifier
event_date	Activity event date
session_cnt	Recorder session count
feature_used	Feature used by customer

03. Transactions
1,014 rows 3 columns

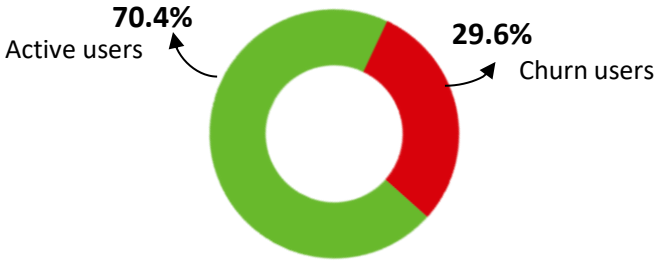
Column	Description
user_id	Customer identifier
trx_date	Transaction date
trx_amt	Transaction amount

04. Status
250 rows 4 columns

Column	Description
user_id	Customer identifier
loan_count	Recorded loan count
avg_trx_amt	Average transaction amount
is_churn	Label churn & not churn

User Base & Churn Rate

The total user base is recorded at **250 users**, with **29.6%** identified as churner (**74 users**) and **70.4%** remaining active (**176 users**)



Rp 3.55 B
Total Transaction Amount

1,014
Transaction Volume

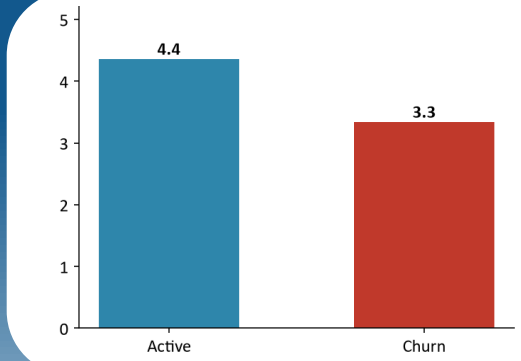
Rp 3.52 M
Average Transaction Amount

Analysis involves 250 users with 1,014 total transaction and 2,959 recorded activity , beside that, the average transaction value is Rp. 3.52 M per transaction, it shows loan/transaction portfolio is significant value

Data Exploration

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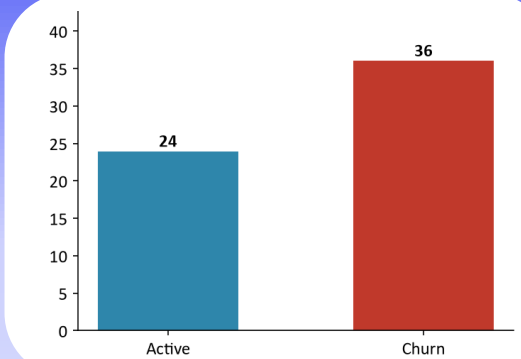
Average number of loan transaction



Interpretation

Churned users took fewer loans than active users (3.3 vs 4.4 on average). Churn is associated with weaker repeat borrowing rather than with a lack of initial uptake.

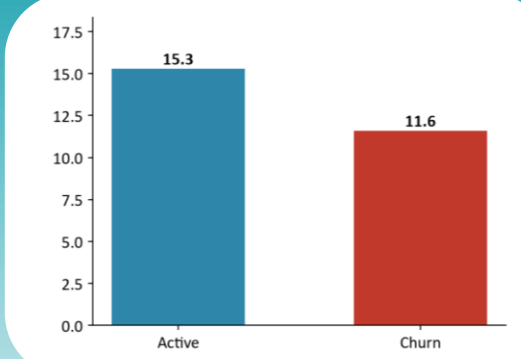
Average of last loan transaction days



Interpretation

Churned users last borrowed 36 days before the data cut-off vs 24 days for active users

Average loan transaction nominal (Rp M)



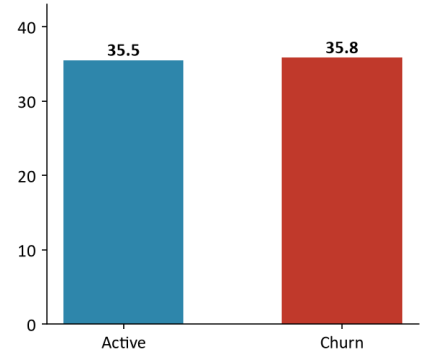
Interpretation

Churned users disbursed Rp11.6M in total vs Rp15.3M for active users

Data Exploration

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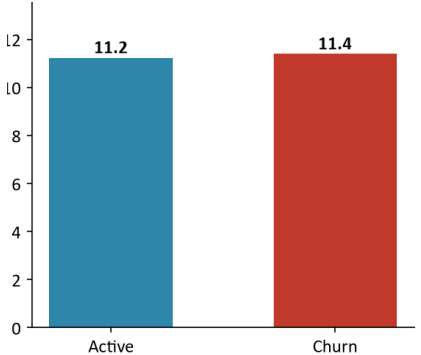
Average session



Interpretation

Churned and active users have almost identical app sessions (35.8 vs 35.5). Session volume does not separate churners, so it is a weak early-warning signal.

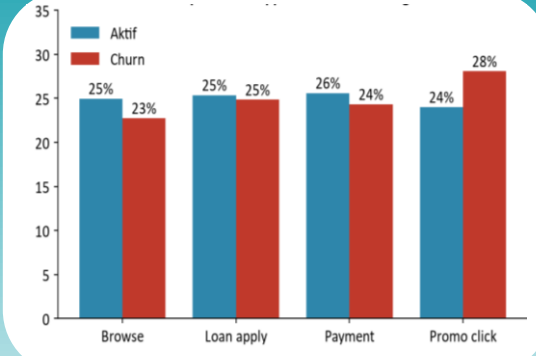
Average active days



Interpretation

Active days are also nearly the same (11.4 for churners vs 11.2 for active users). Churners remain as engaged with the app as active users; the issue lies in borrowing, not app usage.

Proportion of application feature usage



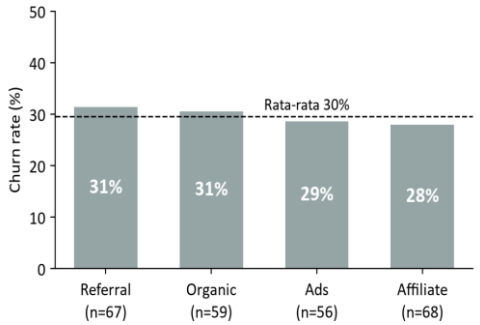
Interpretation

Feature mix is similar for both groups except promo click: 28% of churners' events vs 24% for active users

Data Exploration

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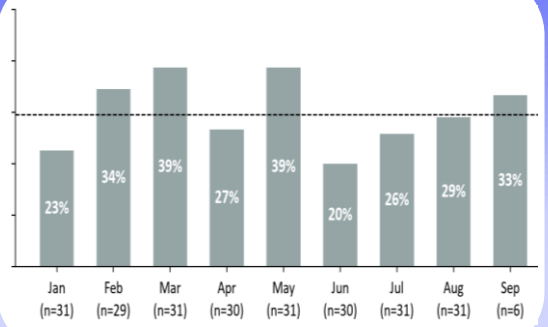
Churn rate per acquisiton channel



Interpretation

Churn ranges narrowly from 28% (Affiliate) to 31% (Referral) around the 30% average, and the differences are not significant. No channel is clearly better or worse at retaining users.

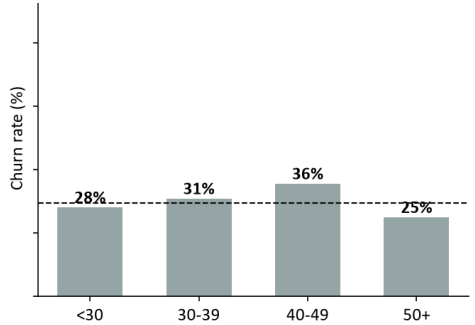
Churn rate per cohort monthly sign up



Interpretation

Churn fluctuates between 20% (Jun) and 39% (Mar, May) without a clear trend, and the differences are not significant. These swings are likely random.

Churn rate per group age



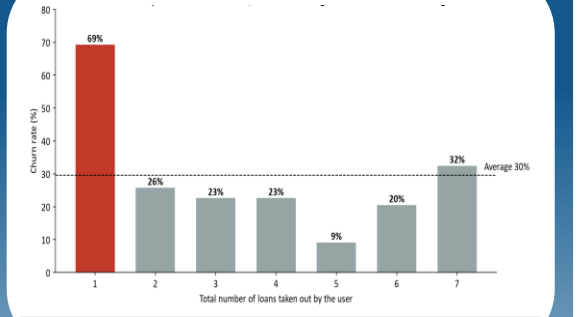
Interpretation

Churn is 25–36% across age groups, highest at 40–49 (36%) and lowest at 50+ (25%), but the differences are not significant. Age does not explain churn.

Data Exploration

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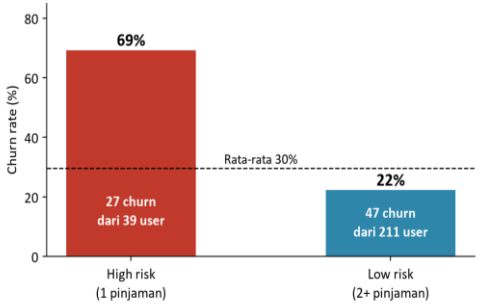
Churn rate per loan transaction



Interpretation

Users with only one loan churn at 69% (27 of 39), more than double the 30% average. Among users with 2–7 loans churn is 9–32% with no clear pattern.

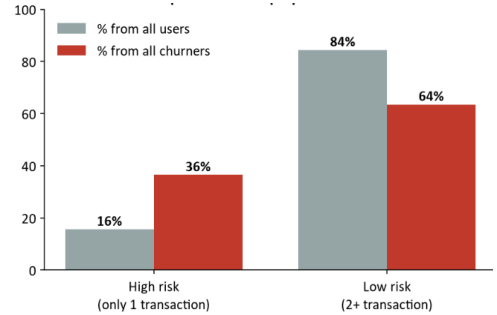
Churn rate per risk segment



Interpretation

High-risk users (only one loan) churn at 69% (27 of 39) vs 22% for low-risk users (47 of 211), around 3x gap. Number of loans works as a simple and effective risk rule.

Proportion of User vs Proportion of Churner



Interpretation

High-risk users are 16% of the base but 36% of all churners (low-risk: 84% and 64%). Targeting them reaches churners 2.3x more efficiently than a blanket campaign.

Churn Prediction — Decision Tree

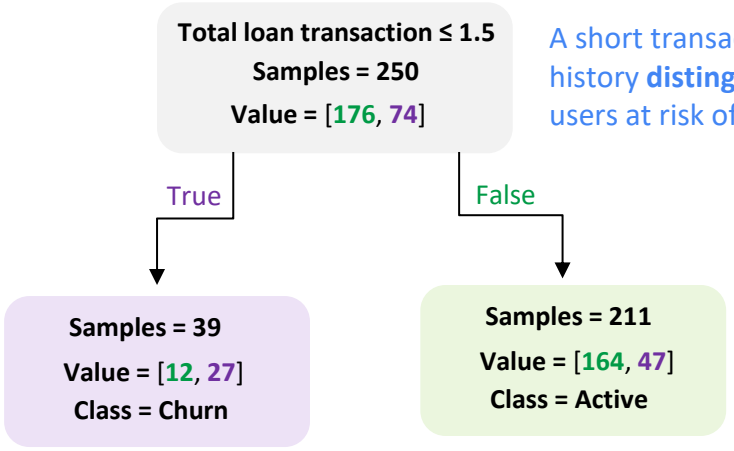
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0.757 ± 0.045	0.672 ± 0.143	0.366 ± 0.116	0.644 ± 0.059
Accuracy	Precision	Recall	ROC-AUC

Confusion Matrix

Ground Truth	Active (0)	Churn (1)
Active (0)	162 TN (True Negative)	14 FP (False Positive)
Churn (1)	47 FN (False Negative)	27 TP (True Positive)

Predicted



A short transaction history **distinguishes** users at risk of churn

The tree selects number of loans as its only split: users with **one loan churn** at **69%** (27 of 39) vs **22%** for users with **2+ loans** (47 of 211). This matches **the risk segmentation**, and **no other feature improved the model.**

Business Recommendations

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Customer retention should be prioritized before scaling acquisition, with the initial focus on converting first-time borrowers into repeat borrowers.

The analysis identifies a high-risk segment of 39 users with only one recorded loan, representing 15.6% of the customer base and approximately 36% of all churners. This creates an opportunity to improve retention through targeted interventions rather than broad-based promotional campaigns.

Retention Strategy

Three initiatives are recommended. First, introduce a **second-loan activation program**, with personalized reminders around day 14 after the first loan and follow-up interventions by day 31. Second, implement a **dormancy early-warning and win-back program** to identify users with prolonged inactivity and engage them through personalized outreach. Third, replace broad promotional incentives with a **behavior-based loyalty program**, rewarding responsible repeat borrowing through eligible benefits. Performance should be monitored using second-loan conversion, reactivation rate, repeat-borrowing rate, and cost per incremental retained customer.

Acquisition and Resource Allocation

Retention should receive initial priority while acquisition continues at its current level. The observed concentration of churn among first-time borrowers supports addressing the existing customer funnel before increasing acquisition spending. Meanwhile, the analysis does not provide sufficient evidence to rank acquisition channels by retention performance. Channel-level budget reallocation should therefore be deferred until stronger evidence is available.

Expected Business Impact

A phased retention pilot should target first-time borrowers and evaluate incremental outcomes against a control group. Success should be measured by improvements in repeat borrowing and reductions in churn, alongside the cost of each incremental retained customer. Because customer acquisition cost, contribution margin, and campaign expenses are not available, the financial return cannot yet be established. Further analysis should also investigate churn among repeat borrowers, who represent a substantial portion of observed churn, to identify additional actionable risk signals

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THANK YOU

robertgunantaa@gmail.com